AI in Financial Risk Modelling

Can foundation models produce valid risk forecasts?

Daniel Traian PELE

Bucharest University of Economic Studies — Dept. of Statistics and Econometrics
Institute for Digital Assets (IDA) · AI4EFin · MSCA Digital Finance
Institute for Economic Forecasting, Romanian Academy

Four ways to produce one number

- Tomorrow's 1% value-at-risk, from four kinds of model.
- A **statistical benchmark** you can compute by hand.
  - Sort the window, read off the quantile. No parameters at all.
- A **time-series foundation model**, pretrained elsewhere.
  - It has never seen this asset, and you cannot inspect what it has seen.
- A **commercial language model**, handed the returns as text.
  - No VaR head, no fitting: the number comes out of prompted generation.
- An **open-weights language model** you run yourself.
  - Three gigabytes, a laptop, and no key. This afternoon, you will.

The question is not which of them is cleverest

- It is whether you can tell, *from the output alone*, that the model answered the question you asked.

Where these two hours go

1. What a risk forecast is, and why 1% is a different problem from 5%
2. The classical answers: historical simulation, GARCH, quantile regression
3. Time-series foundation models: pretraining, zero-shot, and a quantile grid
4. Language models: commercial and open weights, and what they really track
5. Validation: coverage, clustering, loss differentials, contamination, power

The empirical base

- S&P 500, 500 out-of-sample days, 2023-01-04 to 2024-12-30, $\alpha = 1\%$.
- Ten forecasts in the primary comparison, plus three size diagnostics.
- One frozen protocol, five assets in the laboratory.
- Every number regenerated by one script and checked by another.

Act I

What is a risk forecast?

One targets the middle, the other the tail

- A **forecasting** model targets the centre of tomorrow's distribution.

$$\hat{r}_t = \mathbb{E}[r_t \mid \mathcal{I}_{t-1}]$$

- A **risk** model targets a quantile of it.

$$\widehat{\mathrm{VaR}}_t(\alpha) \text{ such that } \quad \mathbb{P}\left(r_t < -\widehat{\mathrm{VaR}}_t(\alpha) \mid \mathcal{I}_{t-1}\right) = \alpha$$

- An excellent R^2 on the centre says nothing about the 1% quantile.
  - They are different summaries of the same conditional law.

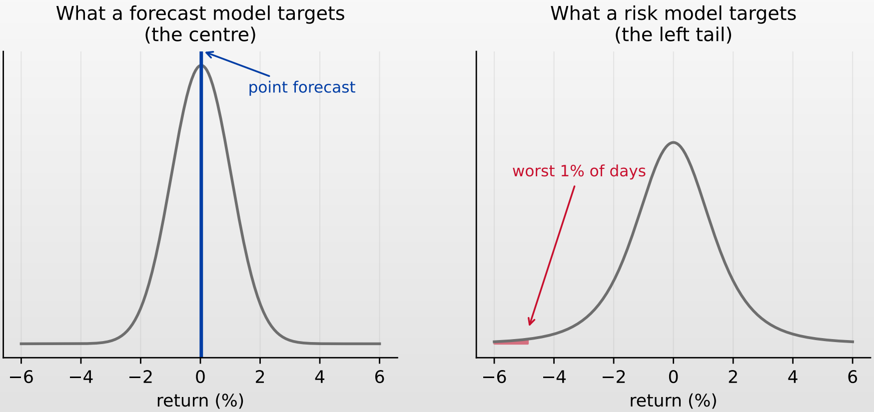
VaR asks how far; ES asks how bad beyond that

- With $L_t = -r_t$ the loss and $q_t(\alpha) = F_t^{-1}(\alpha) < 0$ the return quantile:

$$\mathrm{VaR}_t(\alpha) = -q_t(\alpha) > 0, \qquad \mathrm{ES}_t(\alpha) = \mathbb{E}[L_t \mid L_t > \mathrm{VaR}_t(\alpha)]$$

- Convention for the whole deck: VaR and ES are **positive magnitudes**.
  - Every figure is drawn in **return space**, where the same object is the negative threshold $q_t(\alpha)$.
- VaR is a **threshold**.
  - It says nothing about what lies past it, and it is not sub-additive.
- ES is an **average of the tail**.
  - It is coherent (Acerbi & Tasche, 2002; Artzner et al., 1999), and Basel moved market-risk capital from 99% VaR to 97.5% ES for that reason (Basel Committee on Banking Supervision, 2019).
- We score VaR today.
  - It is the one object every model family here can be asked for directly.

Point forecast, or the whole predictive distribution?



A risk model must deliver a distribution, or at least one quantile of it. Point accuracy is not the currency.

Two factors, and only one of them has data

$$q_t(\alpha) = \sigma_t \times q_\nu(\alpha), \qquad \mathrm{VaR}_t(\alpha) = -\,q_t(\alpha) > 0$$

The scale

- One observation every day.
- Volatility clusters, so it is forecastable.
- A learning algorithm has data to learn from.

The shape

- At $\alpha = 1\%$: **five** events in 500 days.
- The tail index rests on a handful of observations.
- No architecture changes that.

- Every result today is a statement about which of the two a given model got wrong.

Why 1% is a different problem

- 1% is the level supervision counts and the level that hurts.
  - Two or three days a year, and they are the days that end careers.
- It is also where every estimator is at its weakest.
  - A 250-day historical-simulation window holds **two and a half** observations below the quantile.
  - A 500-draw sampling model puts **five** draws below it.
  - A 500-day backtest expects **five** breaches.
- So 1% is not 5% with a smaller number.
  - It is the same question asked where the data runs out, and every method here degrades differently as it gets there.

Act II

The classical answers

The simplest answer: sort the window

- Historical simulation: sort the last W days, read off the empirical quantile.

$$\widehat{\mathrm{VaR}}_t^{\mathrm{HS}}(\alpha) = -\,\widehat{Q}_\alpha(r_{t-W}, \ldots, r_{t-1})$$

- Nothing is fitted and nothing is assumed about the shape of the tail.
  - There is no parametric distribution to mis-specify. Window length and weighting remain modelling decisions.
- Every day in the window carries weight $1/W$.
  - Including one from four years ago, at full weight.
- It therefore **cannot react**.
  - The threshold moves only when an extreme day enters or leaves the window.

AIDA_aug_data

GARCH: model the scale, assume the shape

Engle (1982); Bollerslev (1986)

$$\sigma_t^2 = \omega + \alpha_1 r_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \qquad \widehat{\mathrm{VaR}}_t(\alpha) = -\left(\mu + \sigma_t\, q_\nu(\alpha)\right)$$

- The shock enters **squared**.
  - A -10% day and a $+10\%$ day move σ_t identically.
- The tail **family is imposed**; its parameters are estimated.
  - $q_\nu(\alpha)$ comes from the Student-t assumption, with ν fitted.
  - That is the scale/shape split made explicit in one equation.
- Half a century old, and still the benchmark machine learning must beat.

Quantile regression: learn the quantile end to end

- Minimise the **pinball** (check) loss (Koenker & Bassett, 1978) directly, with q_t the forecast quantile:

$$L_\alpha(r_t, q_t) = (\alpha - 1\{r_t < q_t\})\,(r_t - q_t)$$

- It is **strictly consistent** for the quantile.
  - Minimised in expectation at the truth, which makes it the right thing to rank on — and the breach count is not (Gneiting, 2011).
- Any function class drops in: linear, boosted trees, a neural net.
- The catch is where we are standing.
  - All observations enter the objective, but near 1% identification is weak and the surface is locally flat.
  - A flexible model fits the centre while the tail drifts.

What makes a loss the right loss?

Definition 1 (Consistency)

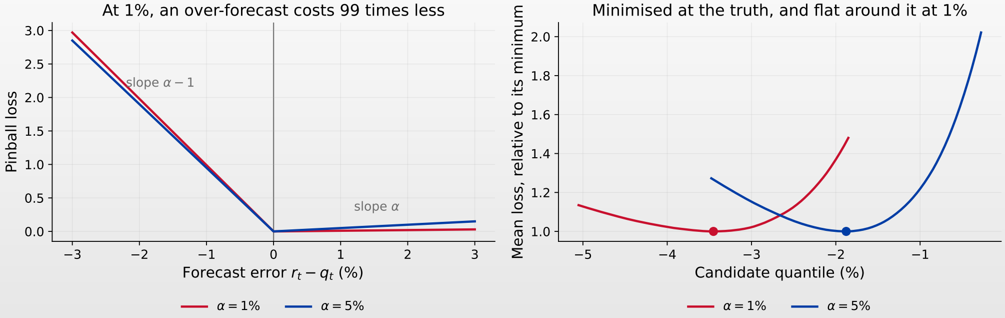
A scoring function L is **consistent** for a functional T if $\mathbb{E}[L(Y, T(F))] \leq \mathbb{E}[L(Y, x)]$ for every F and every report x. It is **strictly** consistent if equality forces $x = T(F)$.

Definition 2 (Elicitability)

T is **elicitable** if some strictly consistent L exists for it. Then $T(F) = \arg\min_x \mathbb{E}[L(Y, x)]$, and models can be ranked by average loss.

- Quantiles are elicitable; the pinball loss does it (Gneiting, 2011).
- ES is *not* elicitable on its own — which is Act V's problem, not this one.

The pinball loss, and why 1% is the hard case



Left: at $\alpha = 1\%$ the two arms have slopes 0.01 and -0.99, so the loss punishes a threshold that is too close far harder than one too far. Right: the mean loss is minimised at the sample quantile, but at 1% every candidate within **0.64** points of it is inside 1% of the minimum — against 0.46 at 5%. Identification is weakest exactly where supervision looks.

Where can AI actually help?

- **The scale.** Thousands of training examples, a well-posed regression.
  - Genuine gains. A small network on log-volatility beats EWMA.
- **The shape.** A handful of events.
  - Flexibility here manufactures confidence, not information.
- **The joint law.** What a pretrained model claims to deliver.
  - Both factors at once, having never seen this asset. That is Act III.

- A flexible model does not create tail observations.
  - It redistributes the ones you have.

Act III

Time-series foundation models

Four words people use interchangeably

- **Supervised forecasting** — fit on *this* series, predict it.
- **Pretraining** — fit once, on a large corpus of *other* series.
- **Fine-tuning** — take a pretrained model, continue training on yours.
- **Zero-shot** — take a pretrained model, run it, change nothing.

Everything in this Act is zero-shot

- No parameter has seen the S&P 500 in a way you control.
- What it saw in pretraining, you cannot inspect. That asymmetry returns in Act V as a validation problem, not a curiosity.

Point forecast, marginal quantiles, or a predictive law?

- Three different outputs, routinely reported as if interchangeable:

1. a **point** forecast $\hat{r}_t$;
2. a set of **marginal quantiles** at fixed levels;
3. a full **predictive distribution** you can integrate.

- VaR needs (2) *at your level*, or (3).
- ES needs (3), or enough of (2) to approximate the tail mean.
- A model that gives you (2) on **its** grid has not given you (2) on yours.
  - Which sounds pedantic until the next slide.

Chronos: the series as a language

- Values are scaled and quantised into a fixed vocabulary of tokens.
- A sequence model is pretrained on a large corpus of time series.
- Forecasting is next-token prediction, decoded back to the value scale.

Two heads, and the difference is the whole Act

- **Chronos-T5** samples paths, so any quantile can be read off.
- **Chronos-Bolt** outputs a *fixed grid* of quantiles, far faster.

Ansari et al. (2024). TimesFM (Das et al., 2024) is the same idea with a different decoder. AIDA_aug_chronos

Why the input is prices

- In the tested pipeline, price **levels** are the compatible input representation.
  - Give it the last 512 closes, ask for the next.
- Recover the implied return:

$$r = 100 \log\left(\hat{P}_t / P_{t-1}\right)$$

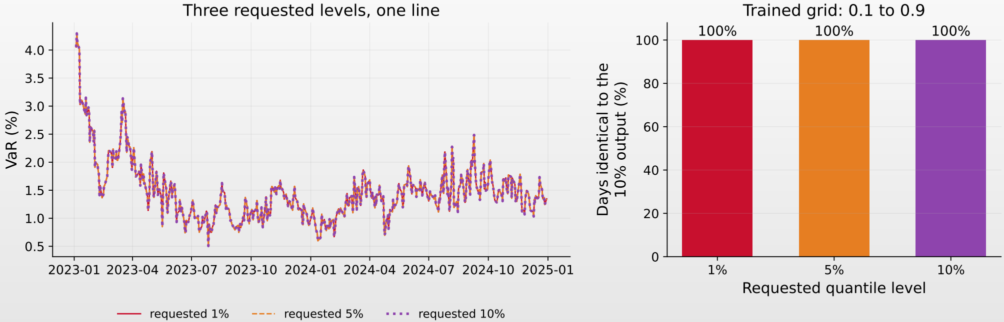
- A monotone transform.
  - So a price quantile maps to the return quantile of the same level.

What happens if you feed returns

- Returns are near-zero-mean noise.
  - Predictive standard deviation **0.14** against a realised **1.22**, and the VaR collapses with it.

Measured in the certified pipeline, S&P 500, 5541 out-of-sample days.

You ask for the 1% quantile. You get the 10%.



Chronos-Bolt was trained on the grid $0.1, 0.2, \ldots, 0.9$. Requests below 0.1 are clamped to the lowest trained level — on **100% of 500 days** the 1%, 5% and 10% outputs are bit-identical.

AIDA_aug_chronos

Why this is the important slide

- A pipeline that requested `quantile_levels=[0.01]` and took the number would have produced:
- a complete set of results, a leaderboard position, and a plot;
- no exception, no failed assertion, no missing value;
- a breach rate around 10% — **ten times** what it claims.

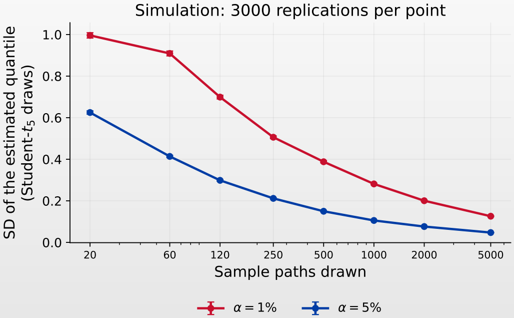
And the natural reading would have been wrong

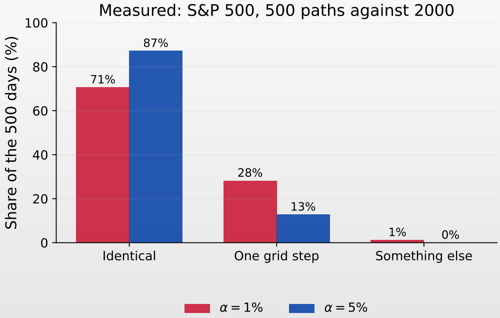
- "The foundation model is badly calibrated in the tail" is a statement about the *pipeline*, dressed as a finding about the model.
- Ask what the artefact *can* answer, not only what you want.
- Then check the values, because the call succeeded.

The sampling head has a floor of its own

- Chronos-T5 draws N price paths and reads the quantile off the sample.
- At $\alpha = 1\%$ with $N = 500$: **five** draws fall below the quantile.
  - The forecast is an order statistic of five points.
- The library default is $N = 20$.
  - Which cannot express a 1% quantile at all: no draw lies below it.
- So the sample count is a modelling choice.
  - With the same standing as the window length in historical simulation, and reported about as often as never.

The same model gives two answers





Right: Chronos-T5 run twice on identical inputs, at 500 and 2000 paths. At $\alpha = 1\%$ the two runs give the **same** VaR on 71% of the 500 days and differ by about **0.63** points on 28% — one step of the token grid. The forecast does not drift with the draw count; it jumps.

AIDA_aug_chronos

Act IV

Language models as risk forecasters

Four different things people mean by "LLMs for risk"

1. **Feature extraction** — sentiment from text, fed to a conventional model.
   - The AI stops at a feature. The risk number is still econometrics.
2. **Event classification** — label the news flow.
3. **Probability elicitation** — ask for $\mathbb{P}$(tail event).
4. **Quantile elicitation** — ask for the VaR itself.

- Only (3) and (4) put the language model *inside* the risk number.
  - That is what we test, following Pele et al. (2026).

Two ways to get a quantile out of a language model

Sampling — the published method

$$\text{serialise} \rightarrow \text{sample } N \text{ completions} \rightarrow \widehat{Q}_\alpha(\text{draws})$$

- Statistically analogous to Chronos-T5 at the sampling-and-quantile stage: the quantile is an order statistic of the draws, and N sets the floor.

Elicitation — what we also run here

$$\text{serialise} \rightarrow \text{ask for } q_\alpha \rightarrow \text{parse the reply}$$

- One call instead of N, and it can be wrong in ways sampling cannot be: a positive quantile, an inverted pair.

Sampling follows Pele et al. (2026) and LLMTime (Gruver et al., 2023); the serialisation is theirs (`base 10, prec 2, signed, ", " separator`).

AIDA_aug_llm

The prompt is a modelling decision

- Four information sets, and the differences between them are the experiment:
- `series` — the last 60 returns, **anonymised**.
  - No ticker, no dates, no price levels.
- `series+state` — the same, plus volatility, drawdown, the HS VaR.
- `dated` — the same returns, plus the asset name and the date.
- `dated+news` — the same, plus the real headlines available at $t-1$.

Why two of them hide the date

- The model's pretraining covers these dates.
  - "S&P 500, 5 August 2024" invites recall, and recall is not available for tomorrow.

One rule decides whether the headlines mean anything

The as-of rule

- A headline may enter the prompt for date t only if it was published at or before the **as-of cutoff** on $t-1$.
  - That cutoff is the asset's own market close.

- The cutoff is measured per exchange, not assumed.
  - S&P 20:00, DAX 15:30, Nikkei 06:00, gold 17:30, Bitcoin 24:00 UTC.
  - New York's close applied to the Nikkei would hand it fourteen hours of news published after Tokyo shut.
- Consecutive cutoffs **tile** the calendar.
  - No article falls in two windows and none falls in none. A fixed 24-hour window silently drops every evening and every weekend.

AIDA_aug_news

What the feed gave us

EODHD, S&P 500 index feed

- 35 293 articles fetched
- 2 128 survive the as-of rule
- 447 of 500 dates covered (89%)
- 4.8 headlines per covered date
- **0** published after their cutoff

The first symbol was wrong

- `SPY.US` returned 222 articles for two years.
- It left **67%** of dates empty.
- The feed was chosen on measured coverage, not plausibility.

- Median publication lead before the cutoff: **1.1 hours**.
  - The text is genuinely fresh, and genuinely prior.

Why the control is `dated`

- Real headlines *reveal the period*.
  - They name companies, events and numbers that place the text in time.

Configuration	period revealed	headlines	role
`series`	no	no	anonymised baseline
`dated`	yes	no	**control**
`dated+news`	yes	yes	**treatment**

- Comparing `series` with a news configuration would confound "the text carries information" with "the model now knows which period this is".
  - Only the last two rows identify the first effect.

What can go wrong, in increasing order of danger

- **Unparseable output.** The model wrote prose. Loud, and cheap to fix.
- **Well-formed nonsense.** A positive 1% quantile.
  - It asserts the worst 1% of days is a gain, and it parses.
- **Inverted tails.** $q_{0.01}$ above $q_{0.05}$.
  - Both numbers well formed, the distribution impossible.
- **Prompt sensitivity.** The same information, reworded, moves the number.
- **Temporal contamination.** The answer was in the training corpus.
- **Version drift.** The endpoint updates; yesterday is irreproducible.

- Only the first announces itself.

What we asked for, and what came back

One JSON object per day, **both levels in one reply**:

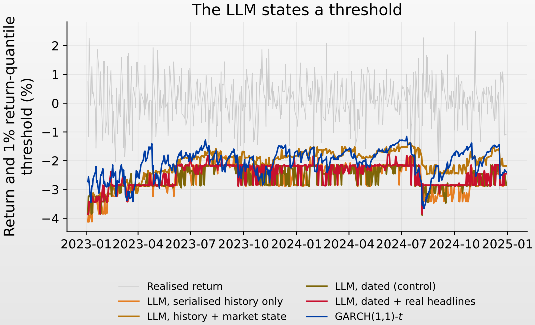
- q01, q05 — the two quantiles
- `prob_negative`, `prob_tail`
- `risk_score`, `confidence`

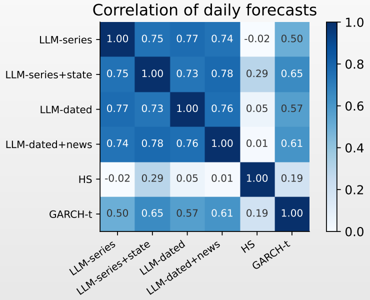
500 days $\times$ 4 configurations

- parsed: 2000 of 2000
- non-negative quantile: **0**
- inverted tails: **0**

- Asking for both levels at once costs a handful of output tokens.
  - Asking twice costs a second 500-request batch, and the two levels then come from two sessions of the model rather than one reply.

The LLM states a threshold, and it moves

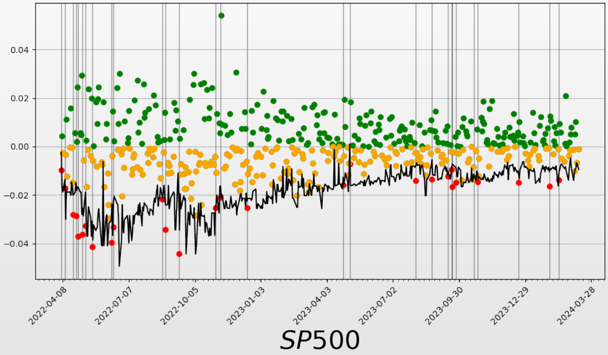




Correlation with GARCH-t is **0.50** for the anonymised configuration, and far lower with historical simulation. This is consistent with the language model primarily tracking volatility — the factor that has data.

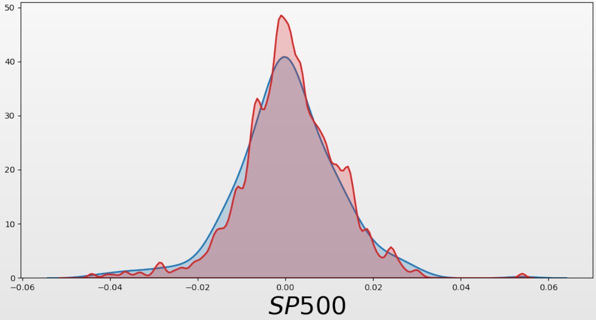
AIDA_aug_llm

The published study, on the S&P 500



Pele et al. (2026), GPT-4, sampling with a 120-day context. Black is the VaR path, green and orange the returns above it, red the breaches. One of nine asset panels in the paper: same prompt, same parser, nine markets — the design this course reruns on open weights.

What the sampled distribution looks like next to the truth



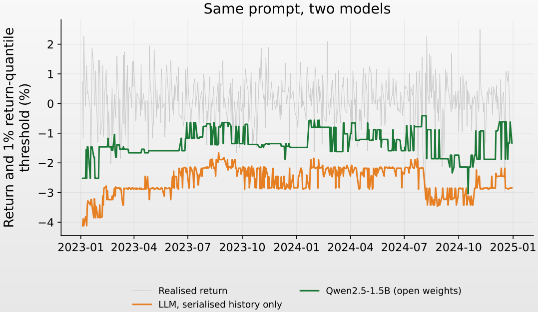
Blue: realised log returns. Red: the language model's draws, pooled over the test span. The draws are more concentrated at the centre than the returns are, and the difference sits in the shoulders rather than the far tail — the same finding as the threshold plot, read off the density instead of the path. Pele et al. (2026), GPT-4, context 120.

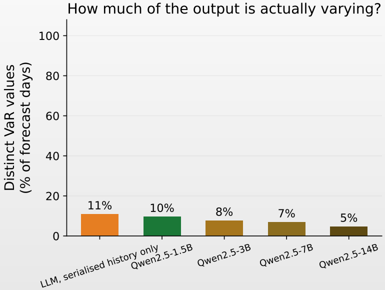
A model you can run on your own laptop

- A laboratory whose central exercise needs a credit card is a demonstration.
  - So the same prompt goes through **open weights**, downloaded from Hugging Face exactly the way Chronos is.
- `Qwen2.5-1.5B-Instruct` (Qwen Team, 2024) — about 3 GB and seconds per forecast, which is what a laptop can run.
  - This is the leg you run yourself this afternoon.
- The package also ships 3B, 7B and 14B, run over the same 500 days.
  - They answer one question: **does a larger model give a more informative VaR?** The next frame is the answer.
- The prompt, the schema and the parser are **imported** from the commercial stage.
  - So the legs differ in exactly one thing: which model reads the prompt.

AIDA_aug_local_llm

The bigger the model, the less it says





The commercial model emits a distinct value on 10.8% of the 500 days; the open models, on 9.5% (1.5B), 7.6% (3B), 6.8% (7B) and 4.6% (14B). Every output is well formed, correctly signed and correctly ordered, and the 14B model moves its VaR on one day in twenty-two.

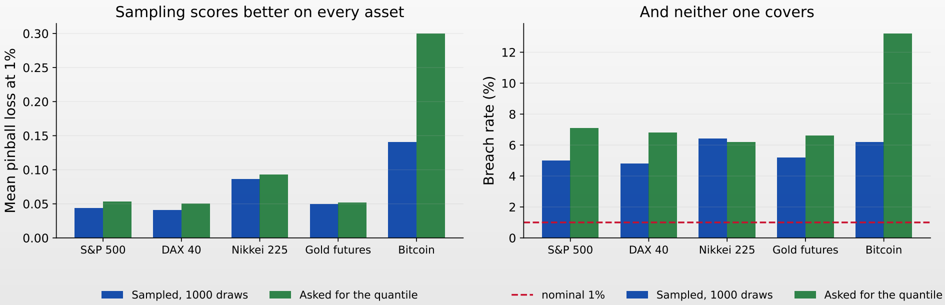
AIDA_aug_local_llm

The check nobody runs: count the distinct outputs

- Every validity test in the pipeline passes on every model in this section.
  - The reply parses, the sign is right and the two quantiles are in order.
- And the forecast moves on fewer than one day in ten.
  - Which no schema, no type check and no assertion can see.
- Only **counting distinct outputs** sees it.
  - One line of code, and it is in nobody's validation checklist.

- "Well formed" and "informative" are different properties.
  - The tooling only measures the first.

The same weights, asked two different ways



One model, two extractions: `Qwen2.5-1.5B` asked for the number, against the same weights sampled **1000** times over the same 500 days, same parser, same scoring. Sampling wins on loss on **all five** assets and emits 15.8% distinct values against 9.5% on the S&P. Kupiec rejects **all ten** runs: breach rates of 4.8% to 13.2% against a nominal 1%.

AIDA_aug_local_llm

How far that comparison goes

- It is the only **fully controlled** experiment in the package.
  - The weights, the days and the parser are held fixed, so the only difference left is how the number was extracted.
- Sampling is the better extraction, and it is not close.
  - Lower loss on every asset, and roughly twice the distinct values.
- Neither is a valid 1% forecast at this model size.
  - Five breaches were expected. The best of these runs delivered 24.
- This is a statement about a 1.5B open model.
  - Pele et al. (2026) ran the sampling method on GPT-3.5 and GPT-4. Nothing here tests those.

Act V

Validation

Backtesting in one line

- Count the days the threshold was crossed:

$$\hat{\alpha} = \frac{1}{T} \sum_{t=1}^{T} 1\left\{ r_t < -\widehat{\mathrm{VaR}}_t(\alpha) \right\}$$

- **Kupiec** tests whether $\hat{\alpha}$ differs from α (Kupiec, 1995).
  - $LR_{UC} \sim \chi^2_1$ under the null.
- **Christoffersen** (Christoffersen, 1998) tests whether the breaches *cluster*.
  - Right count, wrong timing is still a failure: breaches that arrive in one fortnight are the fortnight that matters.
- Their sum is conditional coverage, χ^2_2.

AIDA_aug_backtest

Kupiec and Christoffersen in full

- With T days, x breaches and n_{ij} transitions between breach and no-breach:

Definition 3 (Unconditional coverage)

$$LR_{UC} = -2\log \frac{(1-\alpha)^{T-x}\,\alpha^{x}}{\left(1-\frac{x}{T}\right)^{T-x}\left(\frac{x}{T}\right)^{x}} \sim \chi^2_1$$

(Kupiec, 1995)

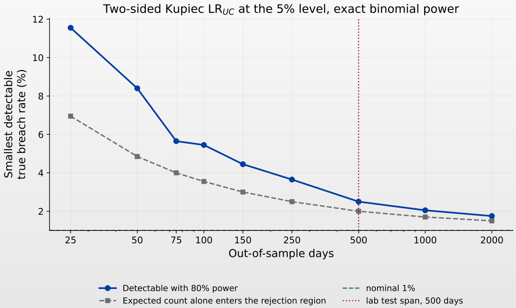
Definition 4 (Independence)

$$LR_{IND} = -2\log \frac{(1-\hat{\pi})^{n_{00}+n_{10}}\,\hat{\pi}^{\,n_{01}+n_{11}}}{(1-\hat{\pi}_{01})^{n_{00}}\hat{\pi}_{01}^{n_{01}}(1-\hat{\pi}_{11})^{n_{10}}\hat{\pi}_{11}^{n_{11}}} \sim \chi^2_1$$

(Christoffersen, 1998)

- $\hat{\pi}_{i1} = n_{i1}/(n_{i0}+n_{i1})$ is the breach probability after state i; $\hat{\pi}$ pools them.
  - Conditional coverage is the sum of the two, χ^2_2.

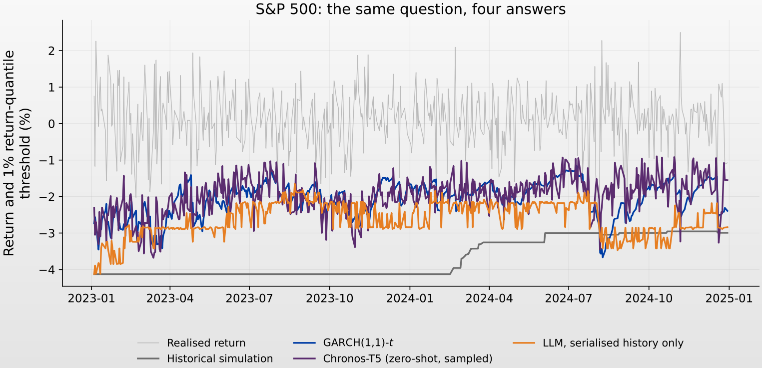
How wrong a model must be before 500 days can see it



Two-sided Kupiec LR$_{UC}$ at the 5% level, exact binomial power. At $\alpha = 1\%$ over 500 days the smallest true breach rate rejected with 80% power is **2.50%**; over 5541 days, 1.45%. The dashed line is the weaker criterion usually quoted — the rate whose *expected* count alone enters the rejection region, 2.00% here — and it is optimistic by half a percentage point.

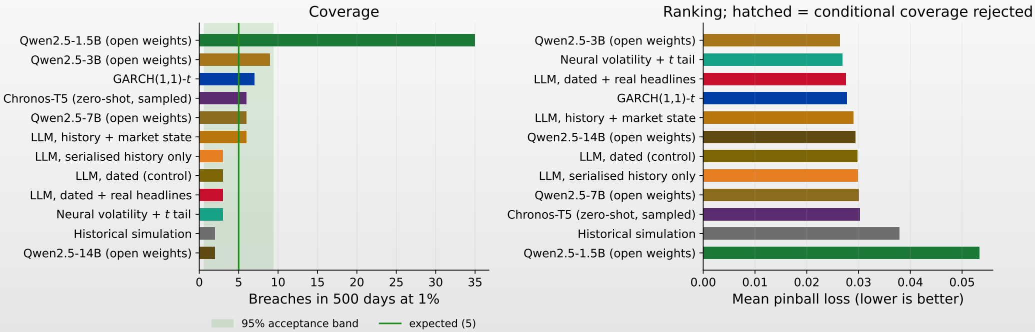
AIDA_aug_backtest

The same question, four answers



S&P 500, 500 out-of-sample days. Same forecast dates, same target, same scoring rule. The information *representation* differs by construction: history length and input form are not the same across HS, GARCH, Chronos and the LLM.

Coverage and ranking, side by side



Expected 5 breaches. Historical simulation delivers 2, GARCH-t 7, and the open-weights model **35**.

AIDA_aug_backtest

What the backtest says

Model	n	Breaches	Rate (%)	UC	IND	Pinball
Neural volatility + t tail	500	3	0.60	pass	pass	0.0269
Claude Haiku 4.5, dated + headlines	500	3	0.60	pass	pass	0.0276
GARCH(1,1)-t	500	7	1.40	pass	pass	0.0277
Claude Haiku 4.5, series + state	500	6	1.20	pass	pass	0.0291
Claude Haiku 4.5, dated (control)	500	3	0.60	pass	pass	0.0298
Claude Haiku 4.5, series only	500	3	0.60	pass	pass	0.0299
Chronos-T5, 500 sample paths	500	6	1.20	pass	pass	0.0303
Historical simulation	500	2	0.40	pass	pass	0.0379
Qwen2.5-1.5B, sampled 1000×	500	**25**	**5.00**	reject	pass	0.0436
Qwen2.5-1.5B, asked	494	**35**	**7.09**	reject	pass	0.0533

- Expected breaches: 5. Ten forecasts, one protocol.
  - Generated by `src/10_slide_tables.py`; source `precomputed/lab_backtest_SPX.csv`.

Eight of ten pass, and that says more about the test

- Breach counts of 2, 3, 6 and 7 all pass Kupiec at 1% on 500 days.
  - Five expected events cannot separate them. The test is not endorsing eight models; it is declining to speak.
- The two rejections miss by factors of five and seven.
  - 5.00% and 7.09% against a nominal 1%. It takes that much to be seen.
- So “passed the backtest” at this sample size means very little.
  - And it is exactly the sentence that appears in model documentation.

- Report the power alongside the verdict, or the verdict is decoration.

Does the S&P result replicate?

Model	S&P 500	DAX 40	Nikkei 225	Gold futures	Bitcoin
Neural volatility + t tail	3 (0.6%)	3 (0.6%)	4 (0.8%)	5 (1.0%)	2 (0.4%)
GARCH(1,1)-t	7 (1.4%)	3 (0.6%)	6 (1.2%)	7 (1.4%)	1 (0.2%)
Historical simulation	2 (0.4%)	0 (0.0%)	5 (1.0%)	7 (1.4%)	0 (0.0%)
Chronos-T5, 500 sample paths	6 (1.2%)	11 (2.2%)	12 (2.4%)	16 (3.2%)	13 (2.6%)
Claude Haiku 4.5, dated + headlines	3 (0.6%)	5 (1.0%)	3 (0.6%)	7 (1.4%)	4 (0.8%)
Claude Haiku 4.5, series + state	6 (1.2%)	7 (1.4%)	7 (1.4%)	11 (2.2%)	7 (1.4%)
Claude Haiku 4.5, dated (control)	3 (0.6%)	4 (0.8%)	3 (0.6%)	4 (0.8%)	5 (1.0%)
Claude Haiku 4.5, series only	3 (0.6%)	3 (0.6%)	3 (0.6%)	2 (0.4%)	8 (1.6%)
Qwen2.5-1.5B, asked	35 (7.1%)	34 (6.8%)	31 (6.2%)	33 (6.6%)	64 (13.2%)
Qwen2.5-1.5B, sampled 1000×	25 (5.0%)	24 (4.8%)	32 (6.4%)	26 (5.2%)	31 (6.2%)
Days scored	500	500	500	500	500
Breaches expected	5	5	5	5	5

Breaches over 500 days, with the realised rate in brackets; red is a Kupiec rejection at 5%. **18 of the 50** model-asset cells are rejected. Both `Qwen2.5-1.5B` rows are rejected on all five markets, and Chronos-T5 passes on the S&P and is rejected on the other four. "Asked" means the model states the quantile in words; "sampled" means it is read off 1000 draws.

AIDA_aug_backtest

Reading that table without fooling yourself

- Fifty tests at the 5% level.
  - Under a true null throughout, about **2 or 3** rejections are expected by chance alone. There are 18.
- The rejections are not scattered. They are concentrated.
  - Ten of them are the two open-weights rows, and four more are Chronos-T5.
- Chronos-T5 is the finding the single-asset table hid.
  - 1.2% on the S&P, then 2.2%, 2.4%, 3.2% and 2.6% elsewhere. One market's pass is not evidence.
- Historical simulation fails in the other direction.
  - Zero breaches on the DAX and on Bitcoin: too conservative to be rejected for being right.

Comparing two models on the same days

Definition 5 (Diebold–Mariano)

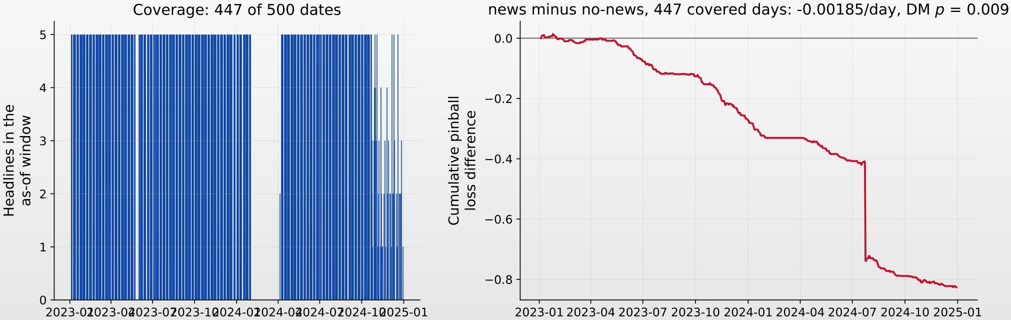
With $d_t = L_\alpha(r_t, q_t^A) - L_\alpha(r_t, q_t^B)$,

$$DM = \frac{\bar{d}}{\sqrt{\hat{\Omega}/T}} \xrightarrow{d} \mathcal{N}(0,1), \qquad H_0 : \mathbb{E}[d_t] = 0$$

(Diebold & Mariano, 1995)

- $\hat{\Omega}$ is a **HAC** estimate (Newey & West, 1987).
  - Loss differentials are serially dependent; the naive variance overstates significance.
- The test is paired, so both models must be scored on the *same* days.
  - A model that skips days is not comparable until the sample is aligned.
- Report $\bar{d}$ and its interval, not only the verdict.

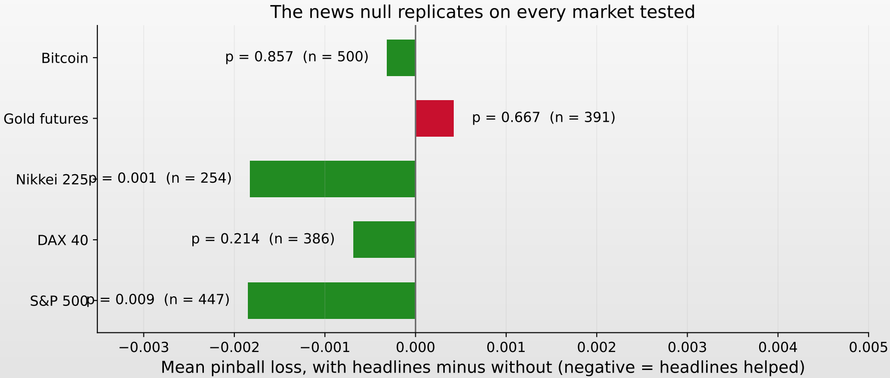
Real news, at the level where it bites



On the 447 scored days carrying at least one headline, the mean pinball loss falls by **0.00185** per day with news, Diebold–Mariano $p = 0.009$. At $\alpha = 5\%$ the same comparison gave $p = 0.61$.

AIDA_aug_news

Does it hold on the other four markets?



Significant on the S&P ($p = 0.009$) and the Nikkei ($p = 0.001$); not on the DAX, gold or Bitcoin.

AIDA_aug_news

How far that result goes

- It is **not** “news improves risk forecasts”.
  - It is: these headlines, at this horizon, at the 1% quantile, on two of five markets.
- **Five tests were run.** Bonferroni puts the threshold at 0.01.
  - The Nikkei survives it at 0.001; the S&P sits on the line at 0.009.
- **Contamination is not excluded.**
  - The prompt hides the date, but the regime is in the corpus regardless.
- **One model, one prompt, one calm period.**
  - 2023–2024 contains no 2008 and no March 2020.

Scoring ES means scoring the pair

Definition 6 (FZ$_0$ loss)

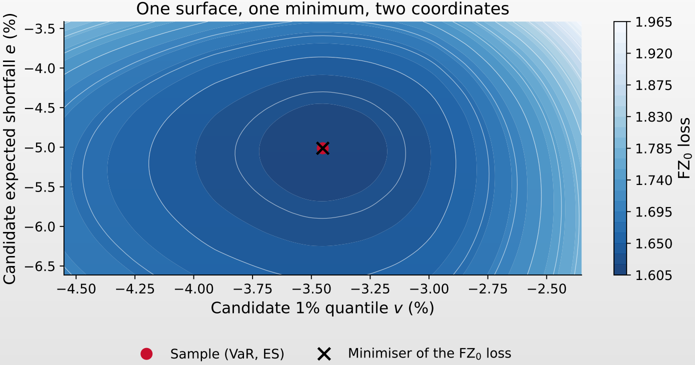
For a quantile $v < 0$ and a shortfall $e < 0$ at level α,

$$L_{\mathrm{FZ}_0}(r, v, e) = -\frac{1}{\alpha e}1\{r \leq v\}(v - r) + \frac{v}{e} + \log(-e) - 1$$

(Fissler & Ziegel, 2016; Patton et al., 2019)

- ES is not elicitable on its own.
  - No scoring function is minimised by the true ES alone, so ES cannot be ranked the way a quantile can.
- The *pair* $(\mathrm{VaR}, \mathrm{ES})$ is jointly elicitable.
  - Which is why regulation can still be backtested (Nolde & Ziegel, 2017).

The minimum sits at the truth in both coordinates



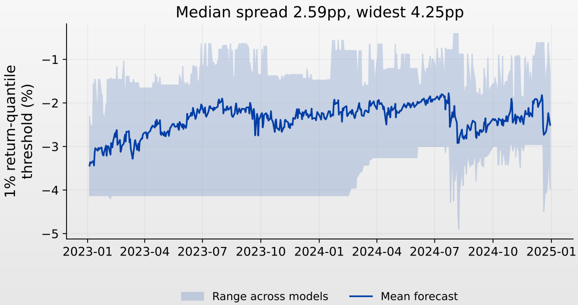
S&P 500 test span. The minimiser of the FZ_0 loss lands on the sample pair in both coordinates, $(-3.45, -5.01)$. The valley is far shallower along ES than along VaR: the tail mean is estimated from the five days below the threshold, and the loss surface knows it.

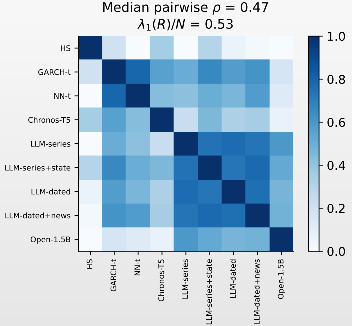
Leakage: the failure that flatters every model

- **Window leakage** — the forecast day inside its own estimation window.
- **Alignment leakage** — a return series shifted one row against dates.
  - Silent, and it inflates every skill measure at once.
- **Selection leakage** — choosing the architecture on the reported period.
- **Temporal contamination** — the pretraining corpus contains the outcome.
  - The first three you can test for. The fourth you cannot, because you cannot read the corpus.

- Anonymising the prompt is therefore a design choice.
  - And the news alignment is audited rather than trusted.

How much of the disagreement is real?





Left: the band between the most and least conservative of the nine, day by day; median spread 2.59pp. A wide band is not evidence of independence, so the right panel measures it: median pairwise $\rho = 0.47$, and $\lambda_1(R)/N = 0.53$ against 0.11 under independence. The four LLM configurations form a block of their own at $\rho = 0.75$.

AIDA_aug_backtest

SYNCRISK: when every desk runs the same model

$$D_t = \max_i \widehat{\mathrm{VaR}}_{i,t} - \min_i \widehat{\mathrm{VaR}}_{i,t}, \qquad \rho_{ij} = \mathrm{corr}\left(\widehat{\mathrm{VaR}}_i, \widehat{\mathrm{VaR}}_j\right)$$

- If every desk runs a similar model, every desk de-risks the same morning.
- Shared **pretraining** is a new channel for that.
  - The same weights behind many vendors, and no way to tell from outside.
- Four prompts are not four opinions: their median ρ is 0.75.
  - Historical simulation is the least correlated with the rest, at 0.21.
- Measuring ρ_{ij} across your own model set is a supervisory question.
  - You compute both quantities this afternoon.

What AI adds to the work of checking

- **You cannot inspect the training data.**
  - Contamination is invisible in principle, not merely in practice.
- **The artefact has properties you did not choose.**
  - A quantile grid, a context length, a tokenisation, a sample count.
- **The endpoint can change under you.**
  - Reproducibility needs a pinned version and a stored prompt — or open weights, which is the other reason they are in this package.
- **Plausible output is the default.** It will always answer.

- None of this argues against using these models.
  - It argues for validating them harder than the ones you fitted yourself.

The laboratory

Two hours in one notebook, and it runs without a key

What you will do this afternoon

- **Run selected forecasts yourself; validate the complete precomputed panel.**

1. **Live, in full:** historical simulation you write, and the whole Chronos-Bolt clamp experiment. Both are seconds per day.
2. **Live, on 5–20 dates:** the Chronos-T5 sampling head and the open-weights language model. Enough to see the mechanism and to check your output against the file.
3. **From `precomputed/`:** the 500-day panels for every model and all five assets.
4. **On the full panel:** backtests, DM tests, the news comparison against its control, and the cross-asset table.

Before you arrive

- Weights are cached in the room. Nobody spends the laboratory downloading 3 GB.
- Your group gets one of the five assets. The ranking is not stable across them, and the reason is the point.

AIDA_aug_lab

Three things to leave with

1. A pretrained model produces a number for any question you ask.
   - Whether it answered *your* question is a separate check, and it is on you. The clamp is the cheap version of an expensive failure.
2. Well formed is not informative.
   - A model can pass every schema check and emit one number all year. Count the distinct outputs.
3. Ranking, validation and power are three different operations.
   - The pinball loss orders models; the coverage tests decide whether any may be used; the power calculation decides whether either statement means anything at all.

⊡ Every number here is regenerated by one script and re-derived from the data by another.

References I

Acerbi, C., & Tasche, D. (2002). On the coherence of expected shortfall. *Journal of Banking & Finance, 26*(7), 1487–1503. https://doi.org/10.1016/S0378-4266(02)00283-2

Ansari, A. F., Stella, L., Turkmen, C., Zhang, X., Mercado, P., Shen, H., Shchur, O., Rangapuram, S. S., Pineda Arango, S., Kapoor, S., Zschiegner, J., Maddix, D. C., Mahoney, M. W., Torkkola, K., Gordon Wilson, A., Bohlke-Schneider, M., & Wang, Y. (2024). Chronos: Learning the language of time series. *Transactions on Machine Learning Research*. https://arxiv.org/abs/2403.07815

Artzner, P., Delbaen, F., Eber, J.-M., & Heath, D. (1999). Coherent measures of risk. *Mathematical Finance, 9*(3), 203–228. https://doi.org/10.1111/1467-9965.00068

Basel Committee on Banking Supervision. (2019). *Minimum capital requirements for market risk* (tech. rep.). Bank for International Settlements. https://www.bis.org/bcbs/publ/d457.htm

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics, 31*(3), 307–327. https://doi.org/10.1016/0304-4076(86)90063-1

Christoffersen, P. F. (1998). Evaluating interval forecasts. *International Economic Review, 39*(4). https://doi.org/10.2307/2527341

Das, A., Kong, W., Sen, R., & Zhou, Y. (2024). A decoder-only foundation model for time-series forecasting. *Proceedings of the 41st International Conference on Machine Learning*. https://arxiv.org/abs/2310.10688

Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics, 13*(3), 253–263. https://doi.org/10.1080/07350015.1995.10524599

Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica, 50*(4). https://doi.org/10.2307/1912773

Fissler, T., & Ziegel, J. F. (2016). Higher order elicitability and Osband's principle. *The Annals of Statistics, 44*(4). https://doi.org/10.1214/16-AOS1439

References II

Gneiting, T. (2011). Making and evaluating point forecasts. *Journal of the American Statistical Association, 106*(494), 746–762. https://doi.org/10.1198/jasa.2011.r10138

Gruver, N., Finzi, M., Qiu, S., & Wilson, A. G. (2023). Large language models are zero-shot time series forecasters. *Advances in Neural Information Processing Systems 36*. https://arxiv.org/abs/2310.07820

Koenker, R., & Bassett, G. (1978). Regression quantiles. *Econometrica, 46*(1). https://doi.org/10.2307/1913643

Kupiec, P. H. (1995). Techniques for verifying the accuracy of risk measurement models. *The Journal of Derivatives, 3*(2), 73–84. https://doi.org/10.3905/jod.1995.407942

Newey, W. K., & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica, 55*(3). https://doi.org/10.2307/1913610

Nolde, N., & Ziegel, J. F. (2017). Elicitability and backtesting: Perspectives for banking regulation. *The Annals of Applied Statistics, 11*(4). https://doi.org/10.1214/17-AOAS1041

Patton, A. J., Ziegel, J. F., & Chen, R. (2019). Dynamic semiparametric models for expected shortfall (and value-at-risk). *Journal of Econometrics, 211*(2), 388–413. https://doi.org/10.1016/j.jeconom.2018.10.008

Pele, D. T., Bolovăneanu, V., Lin, M.-B., Ren, R., Ginavar, A. T., Spilak, B., Andrei, A.-V., Toma, F.-M., Lessmann, S., & Härdle, W. K. (2026). In the beginning was the Word: LLM-VaR and LLM-ES [Code: https://github.com/QuantLet/LLM_Risk]. *Expert Systems with Applications, 295*, 128676. https://doi.org/10.1016/j.eswa.2025.128676

Qwen Team. (2024). Qwen2.5 technical report. https://arxiv.org/abs/2412.15115

Every DOI resolved and every link checked on 2026-08-02. Titles are live links.